



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2022

SUBJECT : CSE 2153 - DIGITAL SYSTEM DESIGN

DIGITAL SYSTEM DESIGN [CSE 2153]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Design a sequential circuit with two JK flip-flops A, B and two inputs E and x. If $E = 0$, the circuit remains in the same state regardless of the value of x. When $E = 1$ and $x = 0$, the circuit goes through the state transitions from 00 to 01 to 10 to 11 back to 00 and repeats. (5)
When $E = 1$ and $x = 1$, the circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00 and repeats.

- 1B) Simplify the function $F(a,b,c,d) = \prod M(1,3,5,8,9,11,15) + D(2,13)$ using K-map to obtain the minimum cost SOP expression and draw its NAND only circuit. (3)

- 1C) List the components in the ASM chart of a Moore type FSM and explain their significance. (2)

- 2A) It is required to design a digital system with the operations shown in Table P.2A

Table P.2A

Operation	Function Performed
I0: Load R0, Data1	$R0 \leftarrow \text{Data1}$
I1: Sub R0, Data2	$R0 \leftarrow R0 - \text{Data2}$

There are four registers R0 to R3 in the system which are connected to the bus through tri state drivers. External data can be loaded into the registers from the Data input, which is connected to the bus using tri state drivers enabled by the Extern control signal. The system also includes a subtractor unit. One of its data inputs is provided by a register A, that is attached to the bus, while the other data input B, is directly connected to the bus. It also has a temporary output register G. The operation is initiated by setting an input w to 1, and the control circuit asserts the Done output when the operation is completed. (5)

i) Draw the block diagram for this system

ii) Determine the timing signals required.

iii) Design the control circuit.

iv) Tabulate the necessary control signals required in different timings.

v) Derive the expressions for the outputs of the control circuit in terms of operations and timing signals

- 2B) Design the least cost implementation with 2:1 multiplexer and other necessary gates for the following function using Shannon's expansion: (3)
 $F(a,b,c,d) = abc + a'cd' + bcd$

- 2C) Construct a Johnson counter which generates 6 timing signals. Write the count sequence and AND gates required for the output. (2)

- 3A) Develop the expressions for the flip flop inputs in the design of a counter with the following repeated binary sequence: 0,1,2,4,5,6,7. Use JK flip-flops. Treat unused states as don't care conditions. Analyze the counter to check if it is self-correcting or not. (5)

- 3B) Distinguish between blocking and non-blocking assignments in Verilog. Draw the circuits for the following codes. (3)

```

module example5_5 (x1, x2, x3, Clk, f, g, h);
input x1, x2, x3, Clk;
output reg f, g;
always @(posedge Clk)
begin
  f = x1 ~& x2;
  g = f ~| x3;
  h = g | f
end
endmodule

```

```

module example5_5 (x1, x2, x3, Clk, f, g, h);
input x1, x2, x3, Clk;
output reg f, g;
always @(posedge Clk)
begin
  f <= x1 ~& x2;
  g <= f ~| x3;
  h <= g | f
end
endmodule

```

- 3C) Illustrate the data transfer between registers via an inter connection network in a computer system. (2)
- 4A) Construct an ASM chart for the pseudo code of a bit counter and explain it. (5)
- 4B) Design a 2-bit Carry Look-Ahead adder using logic gates. (3)
- 4C) Identify the memory element used in sequential circuits and draw the logical diagram of a single input memory element. (2)
- 5A) Construct SR flip-flop using NAND gates. Write its truth table, excitation table. Draw Q_{T+1} waveforms for all combinations of Q_T , S, and R. Avoid forbidden conditions for drawing waveforms. (5)
- 5B) Realize NAND gate using CMOS technology. (3)
- 5C) Design a 3-bit ripple up-counter using positive edge-triggered T flip flop. (2)

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